

PORTFOLIO PROJECT

2024 — 2025

FoodMart Retail Chain

Demand Forecasting
& Inventory Planning

Ibadan, Nigeria | 7 Stores | 20 Products | 18 Months

76,580

Daily Transaction Records

19,262

Total Stockout Events

86%

Supplier Delay Rate

₦13.3B+

Total Revenue Analysed

Project Overview

ABOUT THIS PROJECT

FoodMart operates seven supermarket branches across Ibadan, Nigeria. This project analyses 18 months of transaction data to understand demand behaviour, pinpoint stockout patterns, evaluate supplier reliability, and identify perishable inventory waste risks.

The findings form the foundation for forecasting models and data-driven replenishment decisions.

Demand & Sales Analysis

Inventory Risk

Replenishment Efficiency

Supplier Performance

Expiry & Waste

Demand Trend Analysis

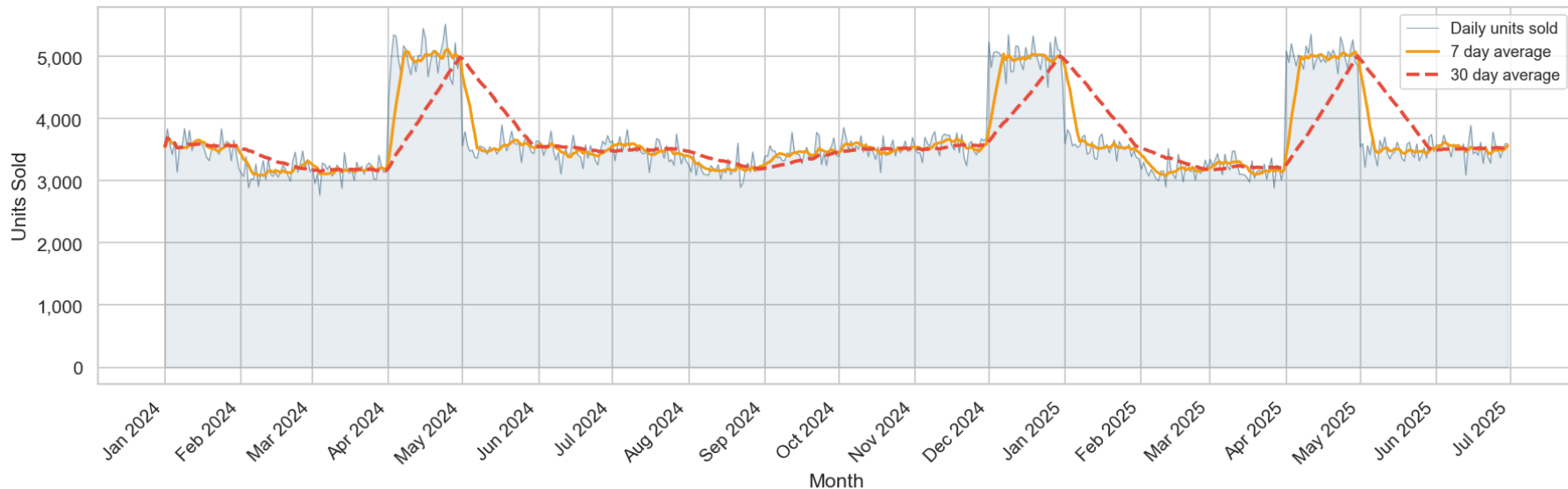
6 Analysis Sections

Tool Stack:

PostgreSQL Python (pandas, matplotlib, statsmodels, Prophet) 18 months of data

Demand Trend Analysis

Total Units Sold Per Day Across All 7 Ibadan Stores



Stable Baseline

3,200 to 3,800 units per day outside seasonal windows

April Surge

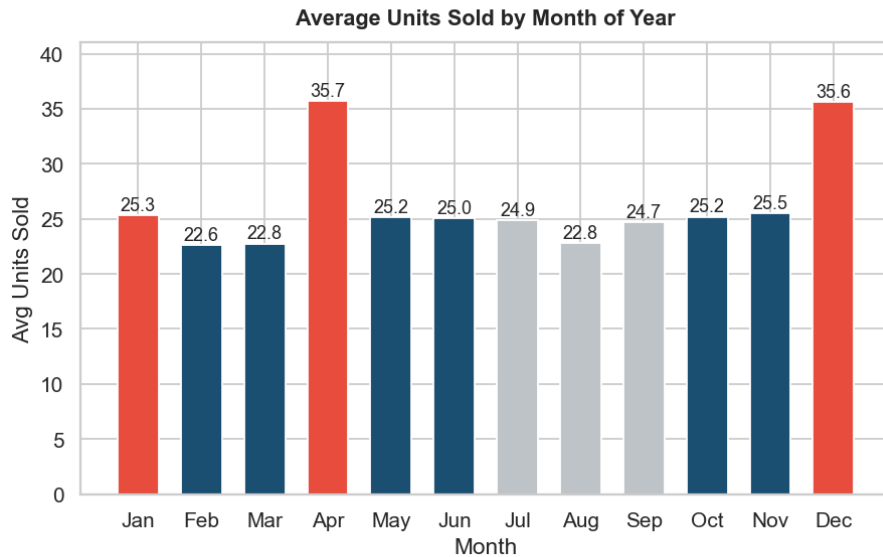
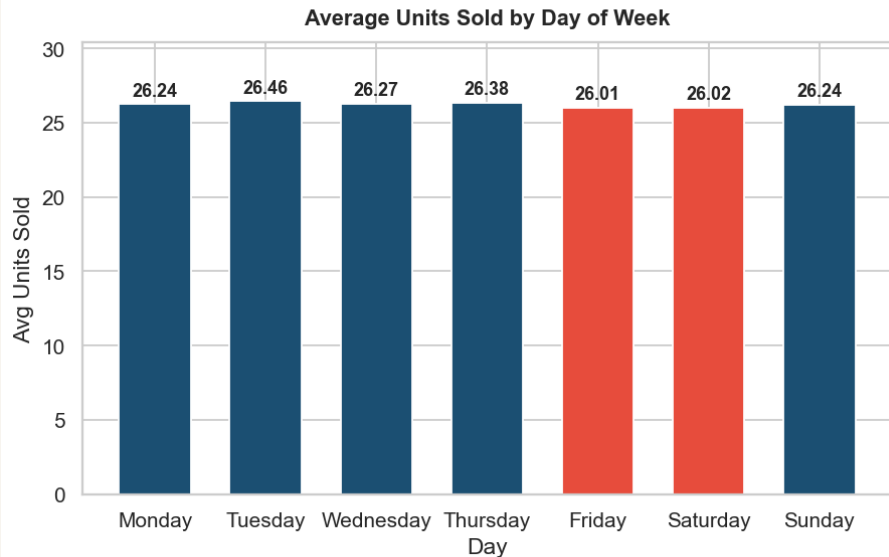
Sales spike to 5,000+ units driven by Easter and Eid seasons

December Peak

Year-end festive period produces a second major demand spike

Weekly & Monthly Demand Patterns

When Are Customers Buying the Most?



Flat Daily Pattern

Daily demand varies by less than 0.5 units on average. No single shopping day dominates, which simplifies weekday replenishment planning.

April & December Dominate

Both months average 35+ units per day versus 22 to 25 units in low months. Seasonal uplift is entirely calendar-driven.

Top Products & Demand Volatility

#1

Bread

142,542 units sold

₦1.12B revenue

Bakery

#2

Indomie Noodles

141,675 units sold

₦1.56B revenue

Food

#3

Rice

140,280 units sold

₦594M revenue

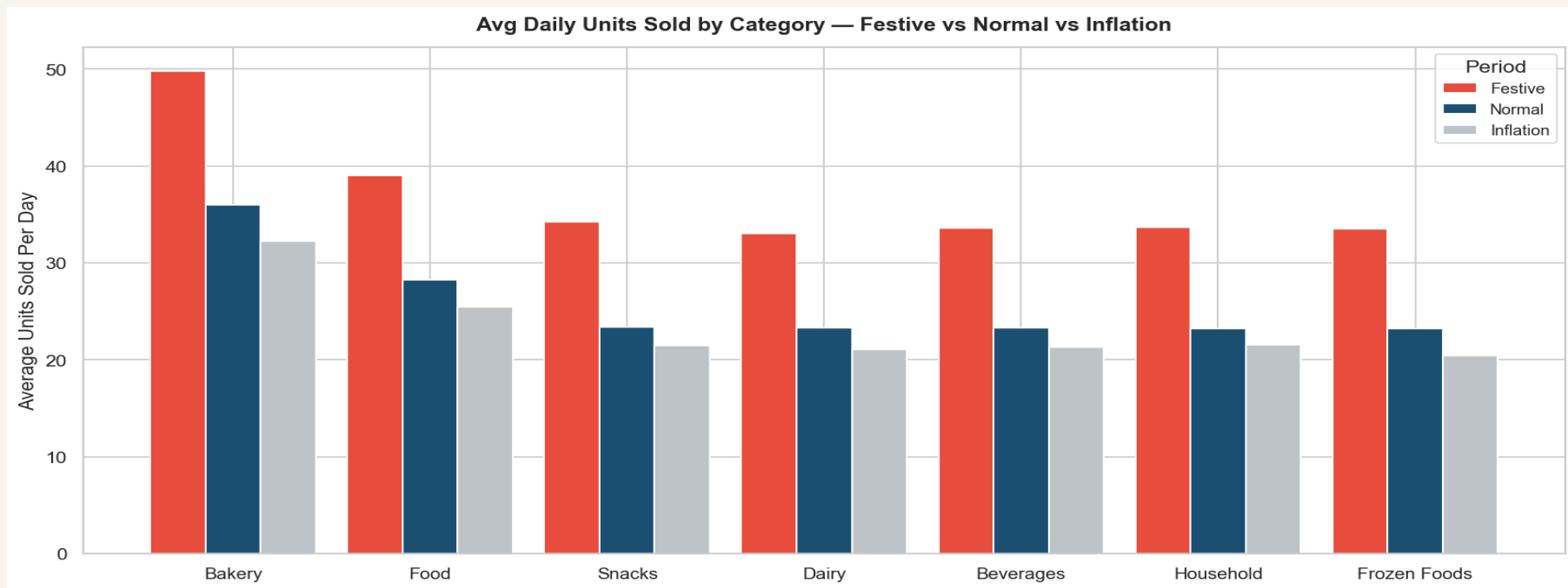
Food

DEMAND VARIABILITY BY CATEGORY



Every category exceeds 50% demand variability, making simple average-based forecasting unreliable.

Seasonal Demand Behaviour



+47%

vs normal demand

Festive Uplift

15-20%

demand reduction across
categories

Inflation Impact

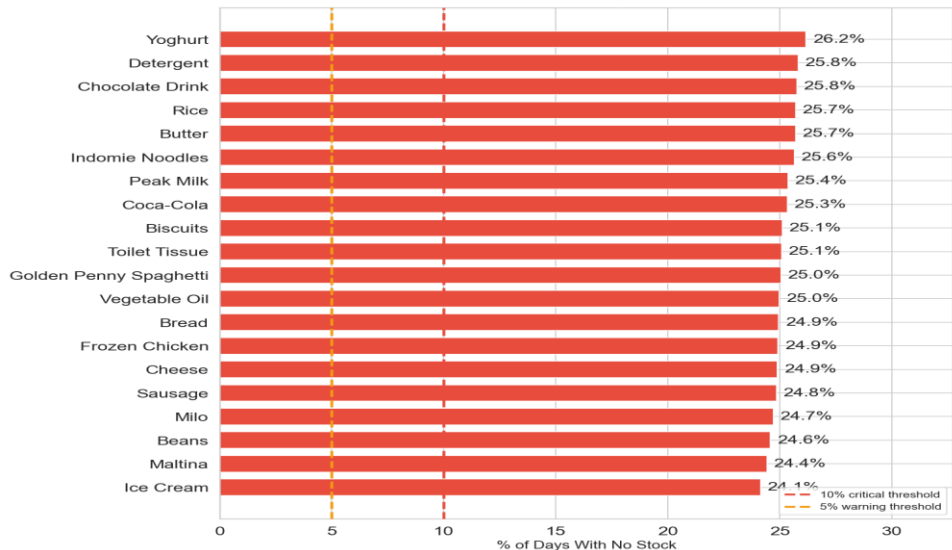
50
units/day
Bakery Peak

highest festive category

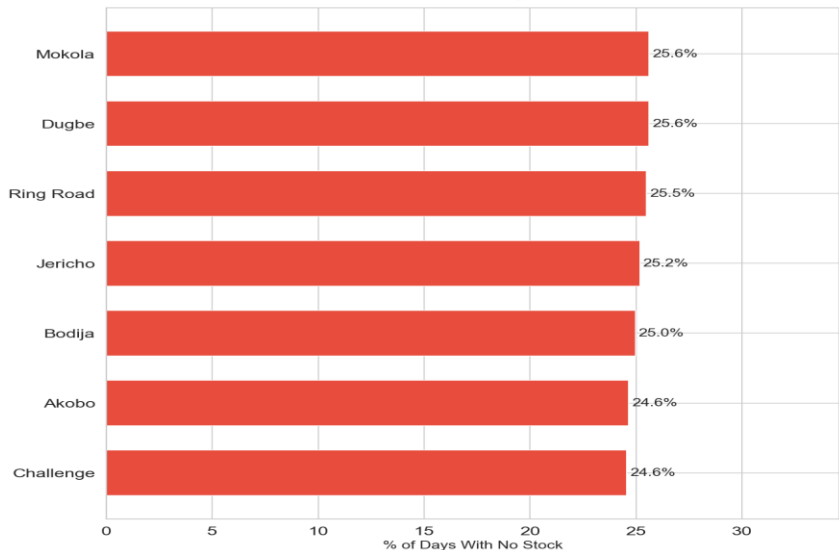
Inventory Risk: Stockout Analysis

Stockout Analysis — Products and Stores

Stockout Rate by Product



Stockout Rate by Store

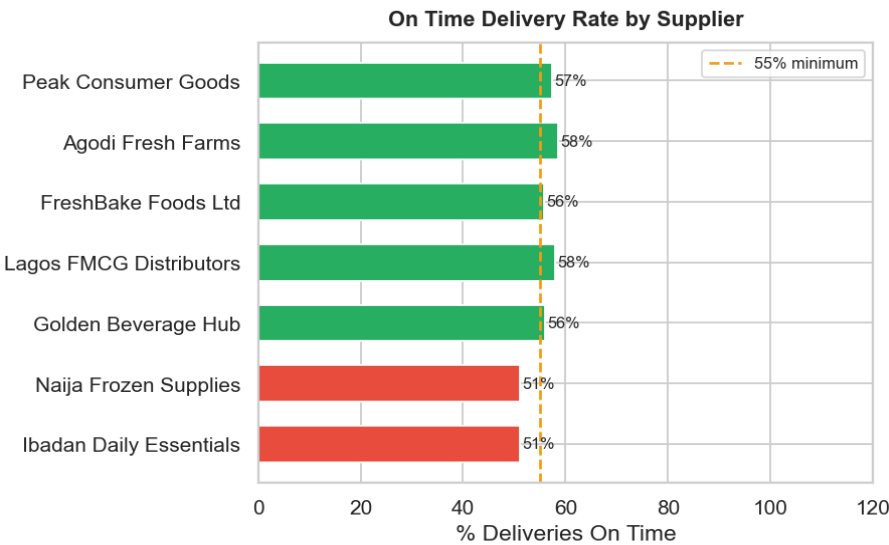


19,262

Total stockout events recorded across all 7 stores and all 20 products over 18 months. Every product in the assortment experienced at least one stockout, and stockout rates range from 24% to 26% across the network, indicating a system-wide inventory planning gap rather than isolated product issues.

Supplier Performance Assessment

Supplier Performance (From vw_supplier_performance)



86.1%

of all deliveries
experienced some delay

4 days

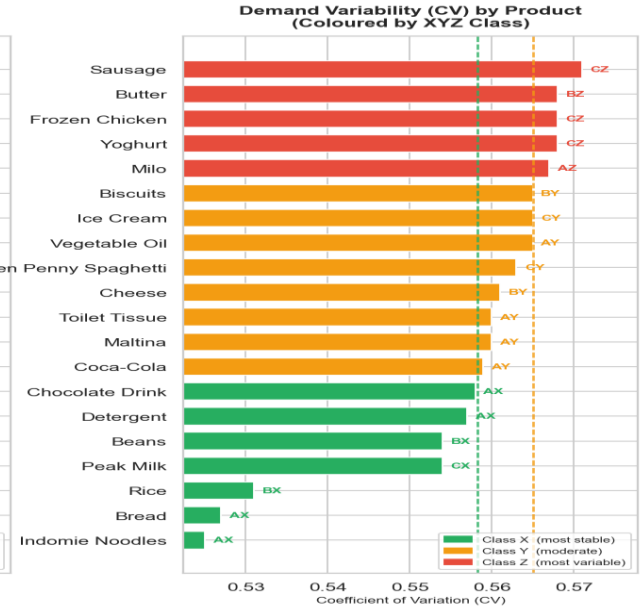
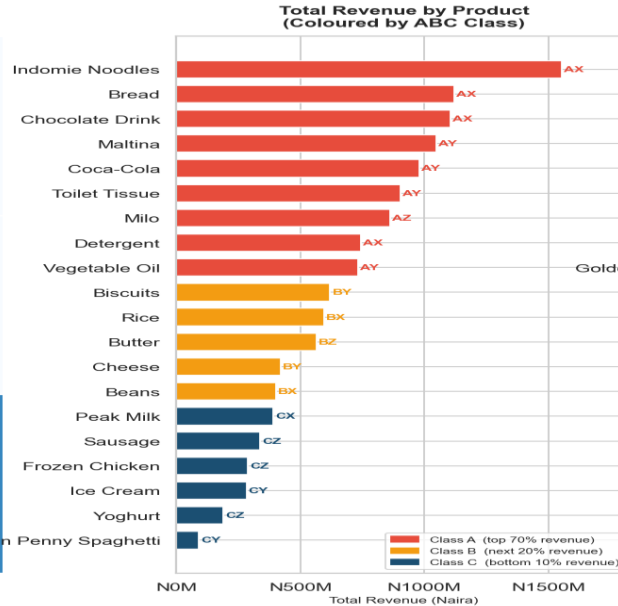
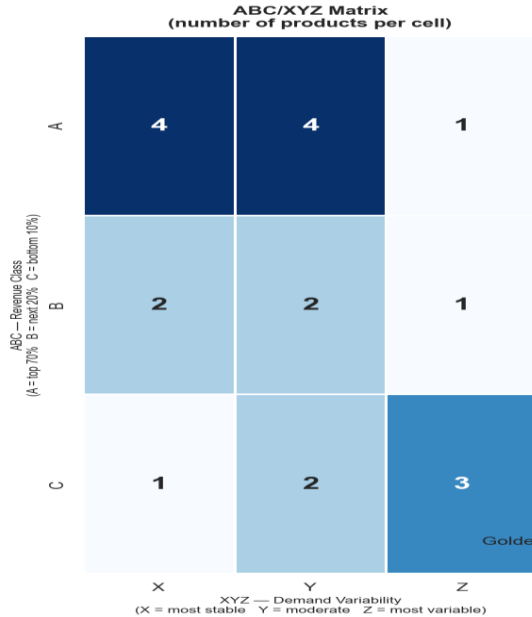
average delay
across all suppliers

51%

lowest on-time rate
(Naija Frozen & Ibadan)

ABC / XYZ Product Segmentation

ABC/XYZ Product Segmentation — Ibadan Supermarket Chain



AX

High revenue, stable demand. Priority planning.

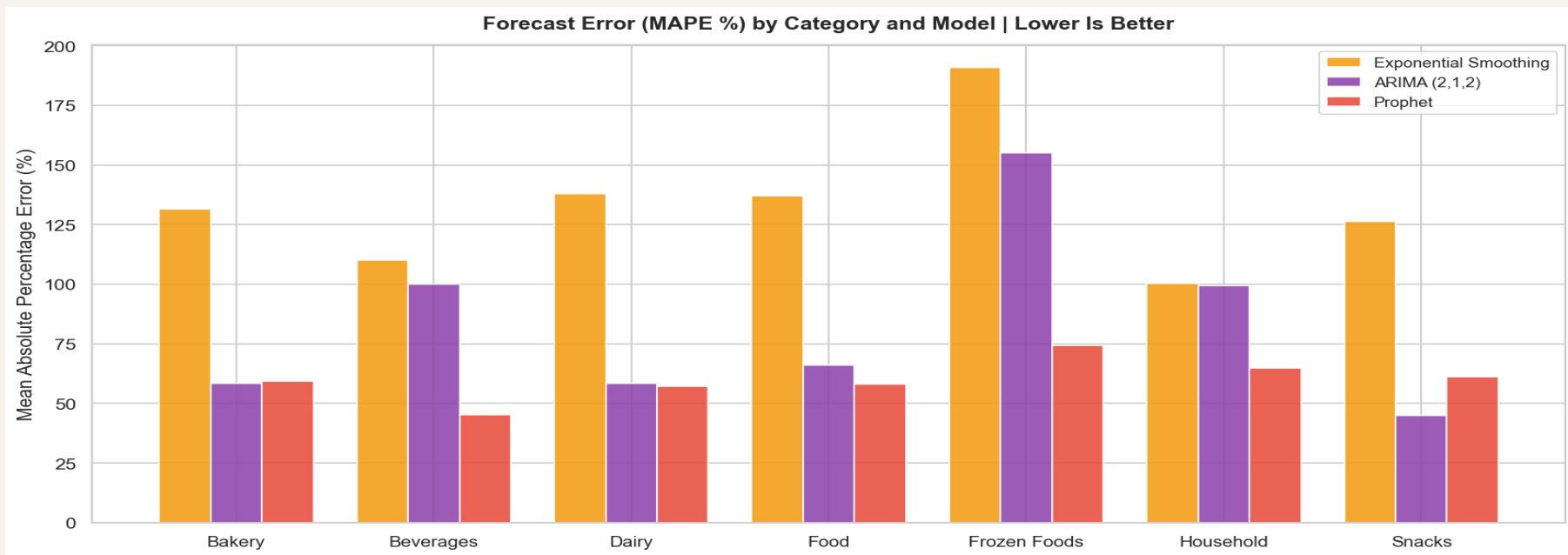
AZ

High revenue, volatile demand. Needs safety stock.

CZ

Low revenue, volatile. Monitor but deprioritise.

Forecast Model Comparison (MAPE)



WINNER

Prophet

Lowest MAPE in 6 of 7 categories. Handles seasonality and holidays natively.

2nd Place

ARIMA (2,1,2)

Strong on stable categories. Competitive with Prophet on Bakery.

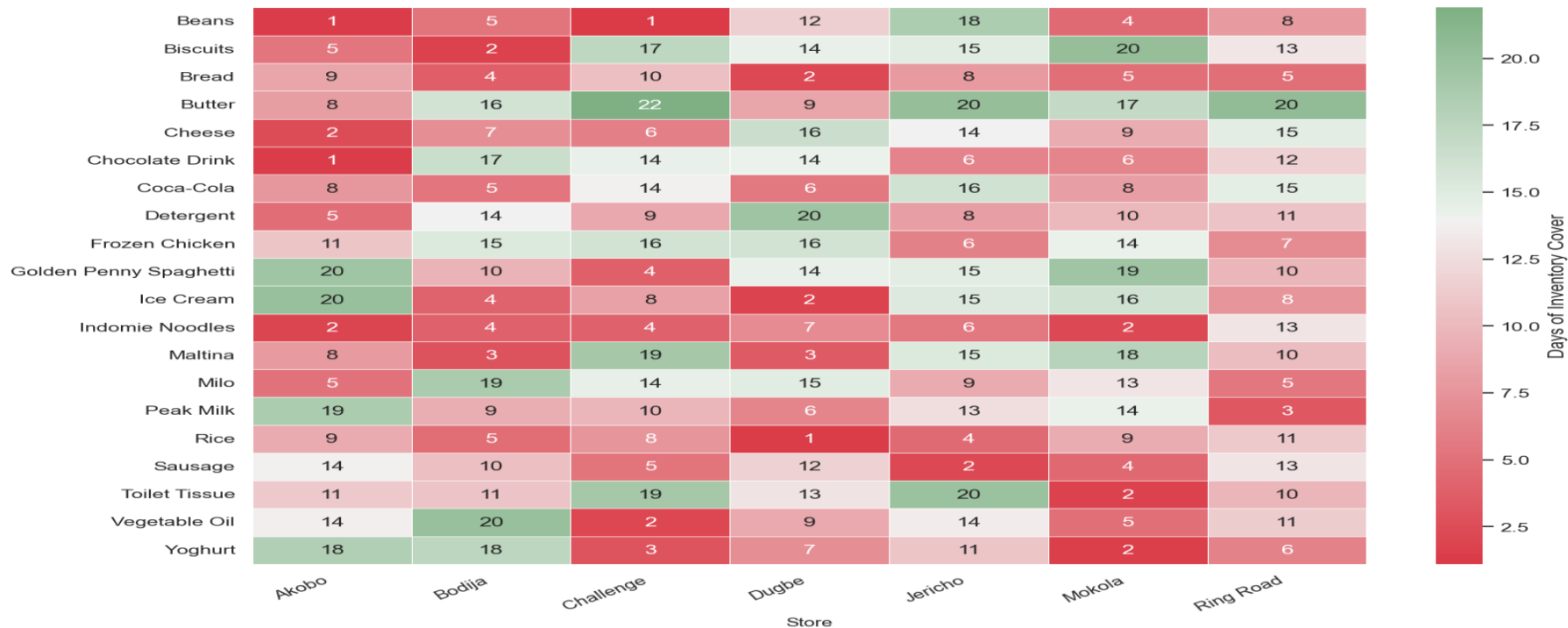
3rd Place

Exp. Smoothing

Consistently highest errors. Fails on trend + seasonal data.

Days of Inventory Cover by Store

Days of Inventory Cover by Product and Store
(Red = critical < 7 days | Green = adequate > 14 days)



Critical under 7 days

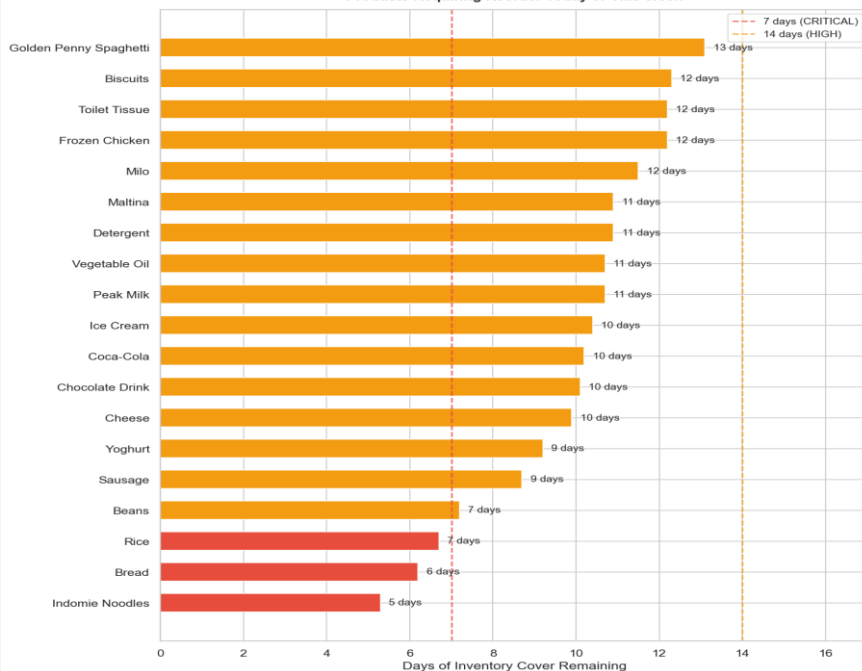


Adequate over 14 days

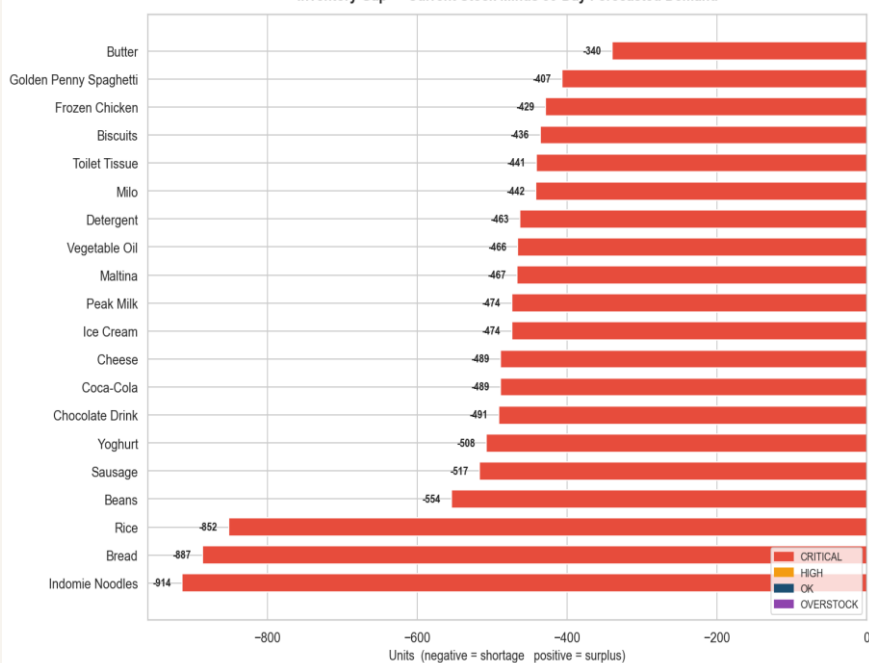
Indomie Noodles, Bread, and Rice are at critical cover levels across multiple stores simultaneously.

Reorder Urgency & Inventory Gap

Products Requiring Reorder Today or This Week



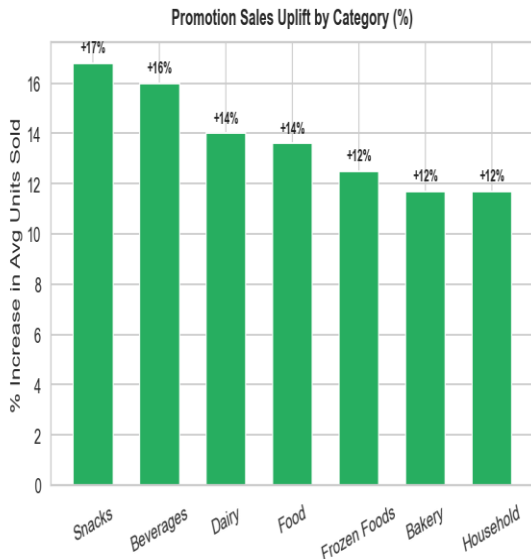
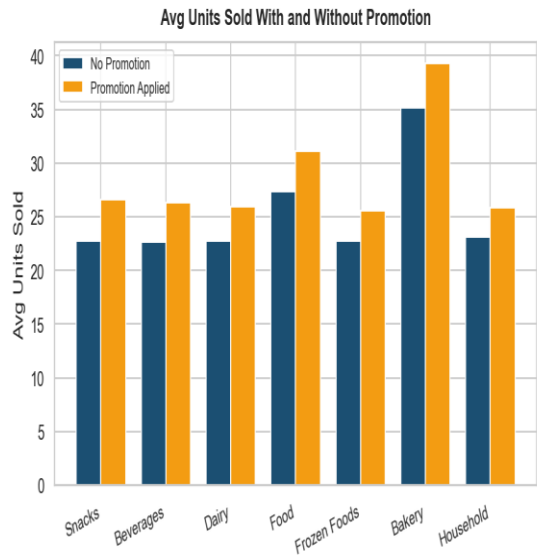
Inventory Gap — Current Stock Minus 30-Day Forecasted Demand



All 20 products show negative inventory gaps versus 30-day forecasted demand. Indomie Noodles (-914 units), Bread (-887 units), and Rice (-852 units) face the most severe shortfalls and require immediate replenishment action.

Promotion Impact & Replenishment Efficiency

Promotion Effect on Demand Across All Categories



PROMOTION FINDINGS

+16%

average uplift across all categories during promotions

Bakery

largest absolute increase: +4.1 units per day

100%

of categories show measurable sales increase with promotions

Replenishment implication: promotional periods consistently require 15 to 20% higher inventory allocations. Forecasting models that ignore promotion calendars will systematically understock during these periods.

Key Findings

01 Demand Is Highly Variable

Every product category exceeds 50% demand variability. Bread, Indomie Noodles, and Rice combine the highest sales volumes with the largest daily swings, making them the hardest to plan.

02 April & December Drive Peaks

Festive periods lift demand by nearly 47% above baseline. Both months appear consistently across both years, confirming these are recurring, predictable events.

03 Stockouts Are Network-Wide

19,262 stockout events occurred across all 20 products and all 7 stores. Stockout rates hover around 25% regardless of category or location.

04 Suppliers Delay 86% of Deliveries

No supplier achieves consistent on-time delivery. Average delays of 4 days across all partners create persistent uncertainty that current replenishment plans do not account for.

05 Prophet Outperforms Other Models

Meta Prophet achieved the lowest forecast error in 6 of 7 categories. Its ability to model seasonality and holiday effects makes it the preferred approach for this retail context.

06 Entire Portfolio Faces Stock Shortfalls

All 20 products show negative inventory gaps relative to 30-day forecasted demand. Indomie Noodles, Bread, and Rice require the most urgent replenishment action.

Recommendations

1

Adopt Prophet for Demand Forecasting

Deploy Meta Prophet as the primary forecasting engine for all seven categories. Its native handling of yearly seasonality and Nigerian public holidays brings error rates significantly below ARIMA and Exponential Smoothing across most categories.

2

Build Seasonal Safety Stock

Pre-position 40 to 50% additional inventory for Bread, Indomie Noodles, Rice, and Bakery products ahead of April and December. Failure to pre-stock these periods results in predictable and avoidable stockouts on the highest-revenue products.

3

Adjust Lead Times for Actual Supplier Performance

Replace planned delivery dates with empirical lead times that add 4 to 5 buffer days to every supplier order. Reorder points must reflect that 86% of deliveries arrive late, not on schedule.

4

Priority Replenishment for Critical SKUs

Indomie Noodles, Bread, and Rice are simultaneously at critical inventory cover levels across multiple stores. These three products represent the highest-value stockout risk in the network and require immediate restocking action.

Thank You

This project analysed 18 months of transaction data across 7 FoodMart stores and 20 products in Ibadan, Nigeria. The analysis covers demand behaviour, stockout patterns, supplier reliability, seasonal effects, and product segmentation.

Forecast modelling using ARIMA, Exponential Smoothing, and Prophet demonstrates that data-driven inventory planning can substantially reduce the 19,262 stockout events observed during the analysis period.

Olajimi Adeleke

Data Analyst | Portfolio Project | PostgreSQL & Python

Tool: PostgreSQL + Python

Data: 76,580 transaction records

Period: January 2024 to June 2025

Coverage: 7 stores, 20 products, 7 suppliers